VIREN VELACHERI

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EDUCATION

Carnegie Mellon University — Pittsburgh, PA
Master of Science in Machine Learning — December 2027
Current Coursework: Introduction to Machine Learning, Probability and Mathematical Statistics, Independent Study (Research)

The University of Texas at Austin — Austin, TX
Bachelor of Science in Computer Science with Honors, Summa Cum Laude — May 2022
GPA: 3.96/4.0 | College Scholar
Relevant Coursework: Neural Networks, Natural Language Processing, Computer Vision, Data Mining, Network Security

WORK EXPERIENCE

Paloa (paloa.ai) — Austin, TX
Founding Engineer (Part-Time) — February 2026 – September 2026
- Engineered a VLM-first basketball video-analysis pipeline (Twelve Labs Pegasus, structured outputs, scoreboard corroboration) on Modal workers, producing player identification, box scores, and AI game summaries.
- Designed a layered player-identification system combining RF-DETR detection, ByteTrack tracking, jersey-OCR VLMs with majority-vote consensus, and SigLIP embedding clustering for team assignment.
- Developed Paloa BENCH, a regression-testing framework scoring outputs against 469 ground-truth shots across 4 games on five dimensions, with pass/fail gates that flag accuracy regressions and account for LLM non-determinism.
- Architected a QLoRA fine-tuning and evaluation pipeline for shot detection (Qwen-VL, Modal A100 GPUs), raising shot-detection accuracy from 79% to 87% with a human-in-the-loop workflow that captures coach corrections.

Cvent, Inc. — Austin, TX
Software Engineer 2 — August 2024 – July 2026
- Architected a cross-service optimization cutting latency by 35% and built Datadog monitoring that halved incident recovery time.
- Doubled team productivity by integrating AI development frameworks into engineering workflows.
- Selected by management to join a two-person, principal engineer-led team for a feature initiative impacting 10,000+ users.

Software Engineer 1 — August 2022 – August 2024
- Engineered React and Redux components that improved dashboard responsiveness by 15%.
- Designed RESTful APIs serving 1,200+ clients with WebSocket handling for 500+ messages per second.

RESEARCH PROJECTS

Measuring Values in Large Language Models
Graduate Student Researcher, Human Sensing Laboratory, Carnegie Mellon University — September 2026 – Present
- Developing a metric to audit how large language models internally represent positions on value-laden social and political issues.
- Evaluating the metric's consistency across varied prompts and contexts through experiments on open-source models.

RAG Retrieval Benchmarking & NER Model Evaluation
Independent Research with Dr. Anshumali Shrivastava, ThirdAI, Inc. — December 2024 – July 2025
- Designed single- and multi-document RAG retrieval benchmarks with custom datasets to standardize retrieval evaluation.
- Evaluated 8 transformer NER models on precision, recall, F1 score, and sub-10-millisecond latency to guide model selection.
- Deployed self-hosted vision-language models for document processing and authored reports informing architecture decisions.

Network Optimization for 5G & Starlink Deployments
Independent Research with Prof. Chandrajit Bajaj, The University of Texas at Austin — February 2024 – July 2024
- Compared genetic algorithms, deep RL, and GNNs for dynamic network optimization; proposed a hybrid RL and GNN approach.

ENTREPRENEURSHIP & PROJECTS

SiliconMCP (siliconmcp.com), formerly LoopCell — Austin, TX
Co-Founder & CEO — July 2025 – September 2026
- Architected Model Context Protocol (MCP) infrastructure for AI-driven chip design: evaluation-as-a-service for LLM-generated SystemVerilog, agentic RTL and testbench generation pipelines, and a pilot with a leading frontier model company.

Resume-Bot (resume.llmchat.ai) & HireBot (hire.llmchat.ai)
Founder — March 2024 – December 2024
- Engineered a provisional-patent LLM system turning resumes into interactive chatbots and an LLM job-candidate ranker.

SKILLS

ML/AI: PyTorch, TensorFlow, Hugging Face, LoRA/QLoRA, MLflow, YOLO, scikit-learn, Computer Vision, NLP
LLM Systems: LLMs, VLMs, RAG, Agentic AI, Model Context Protocol (MCP), LLM Evaluation
Programming Languages: Python, Java, JavaScript/TypeScript, SQL, C/C++, R, MATLAB, SystemVerilog
Frameworks & Tools: React, Node.js, FastAPI, Django, Docker, AWS, Modal, Supabase, MongoDB, PostgreSQL, Spark, Git

HONORS

Cvent "Pat on the Back" Award (2024) | President's Award for Global Learning Finalist (2020)